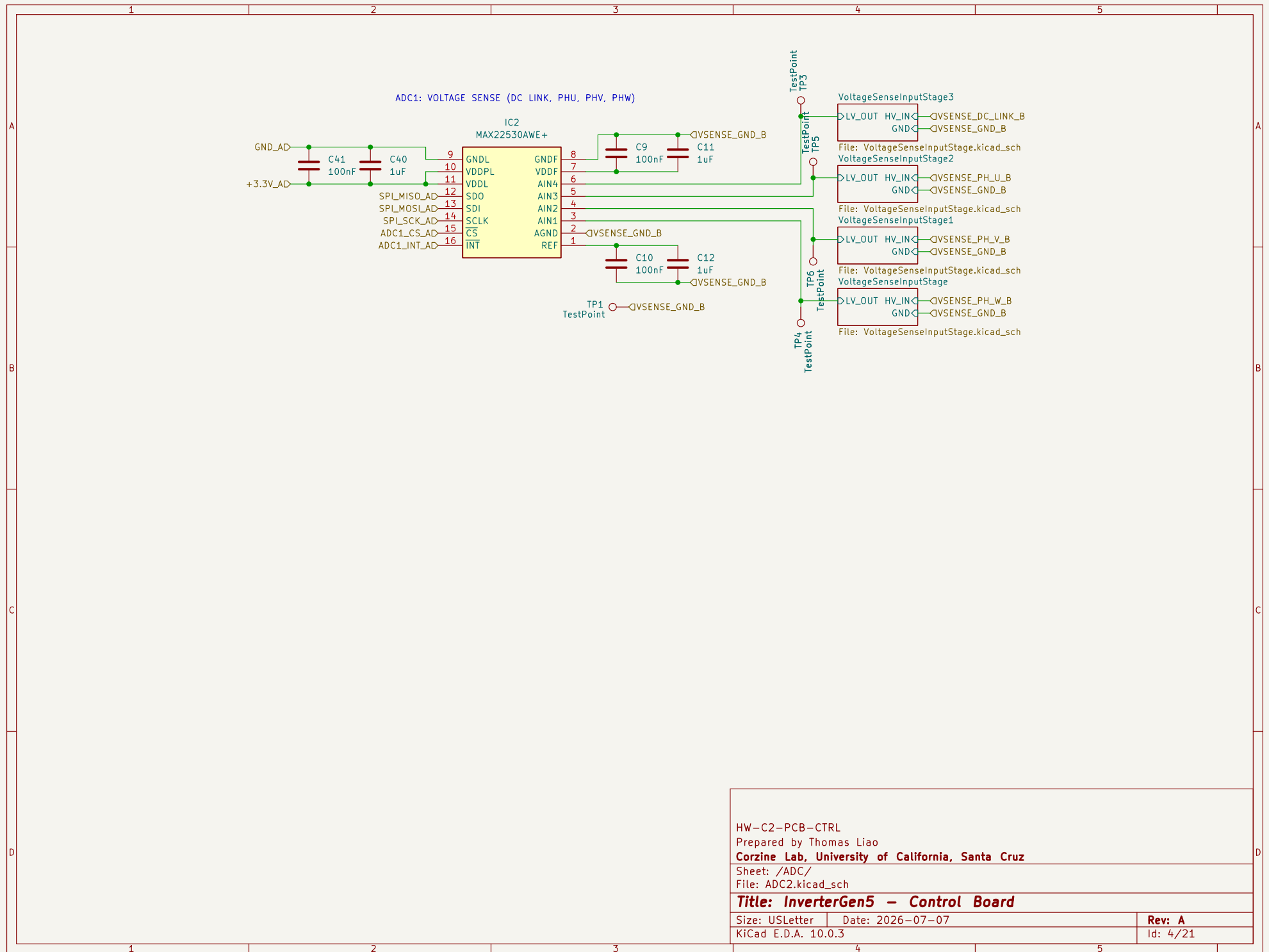


NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)



Rev: A
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HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

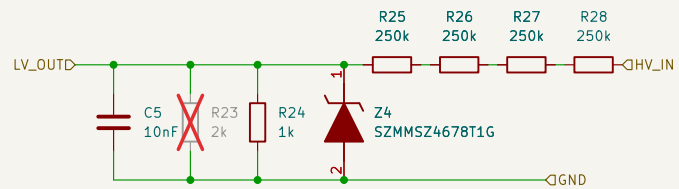
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

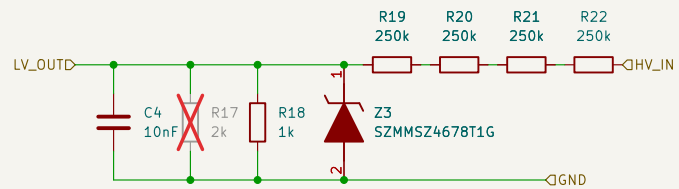
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

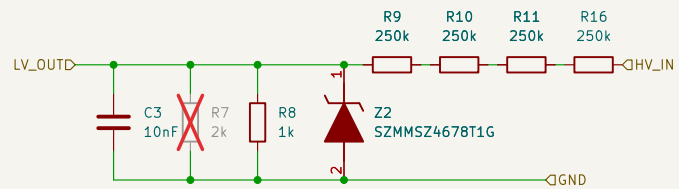
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

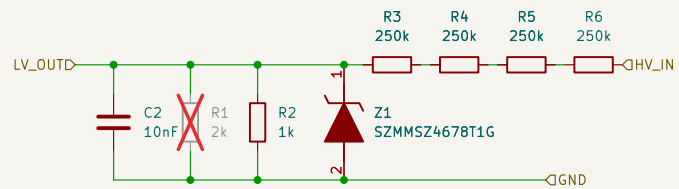
Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

Title: InverterGen5 - Control Board

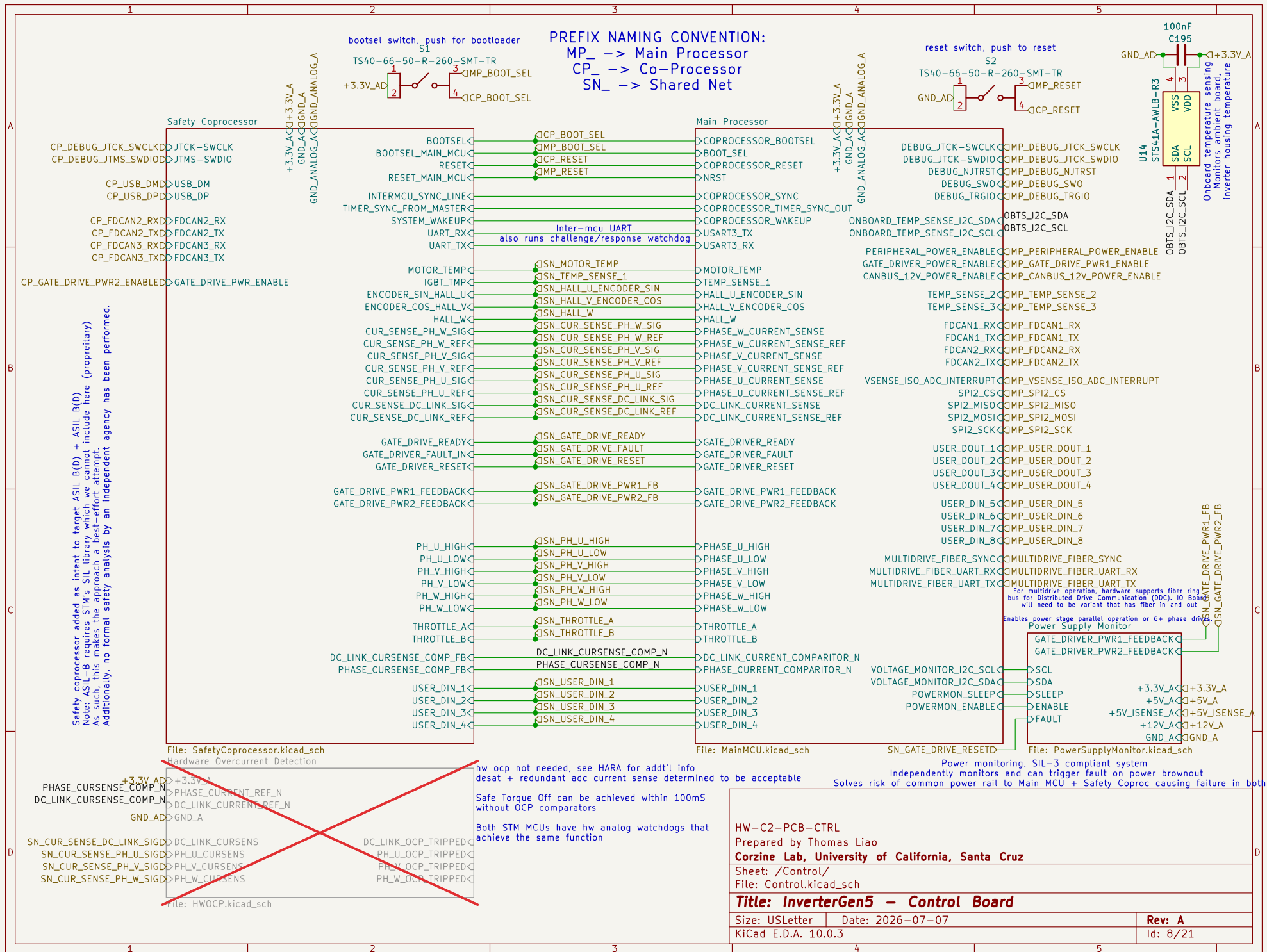
Size: USLetter

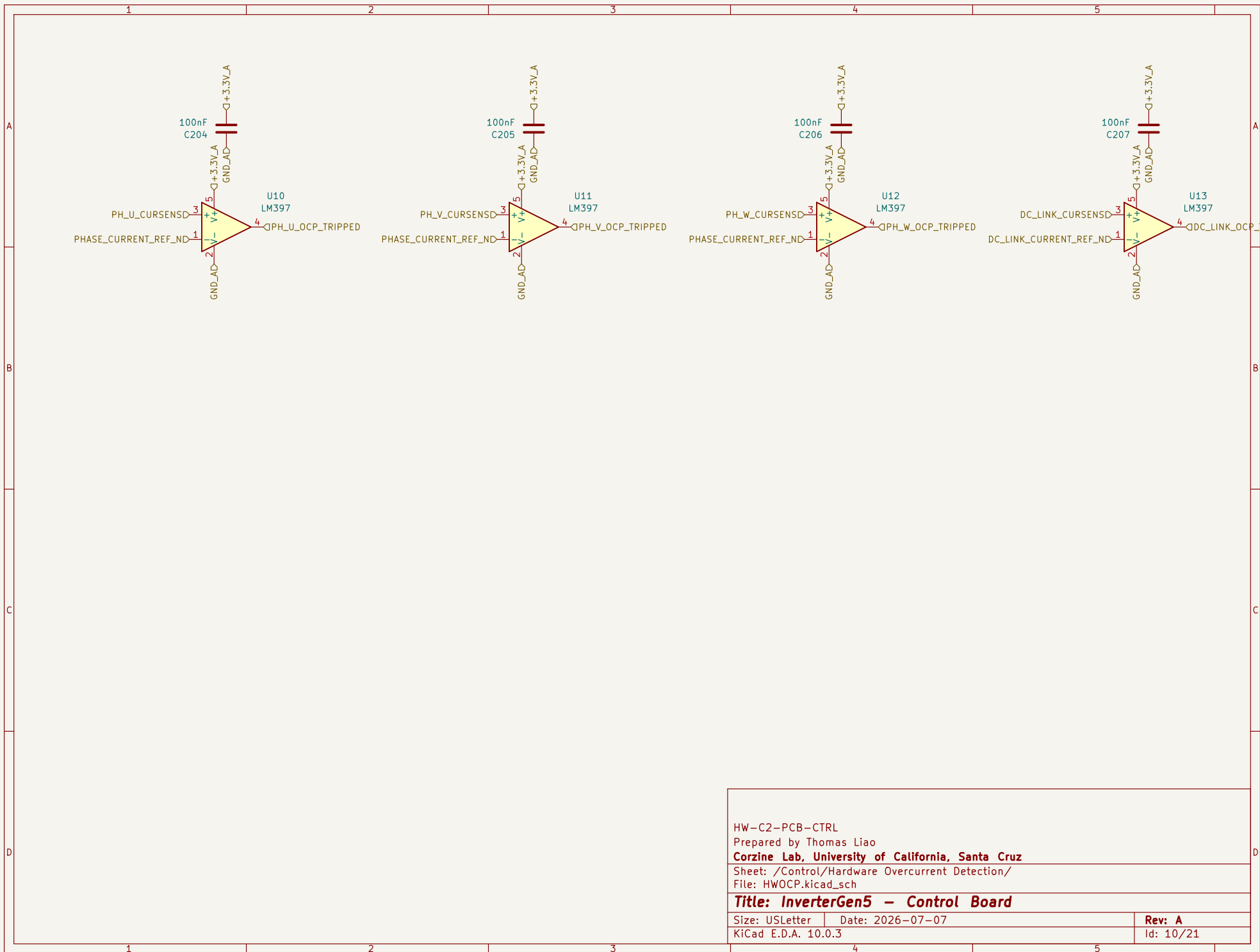
Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

Id: 6/21





HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

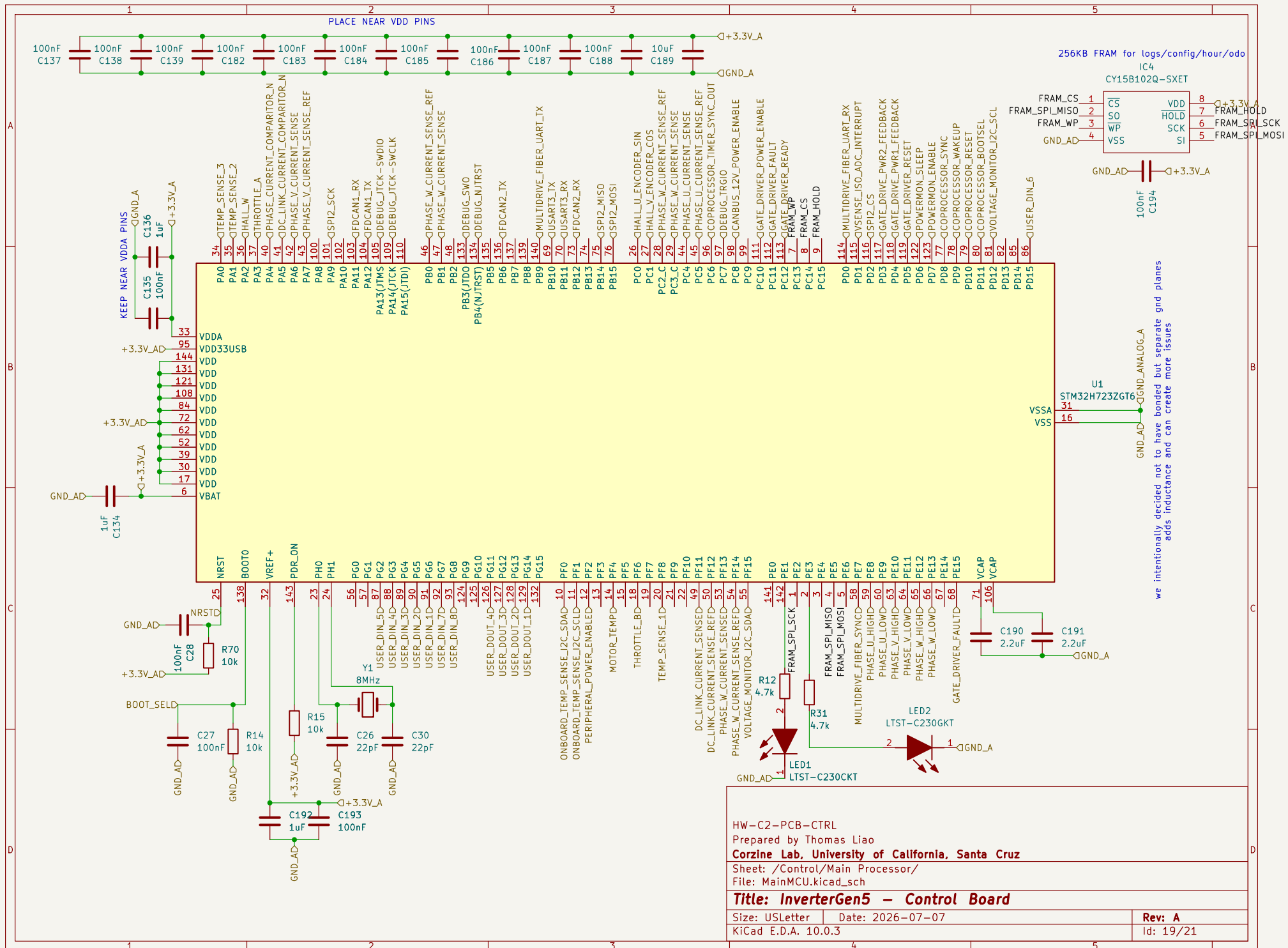
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

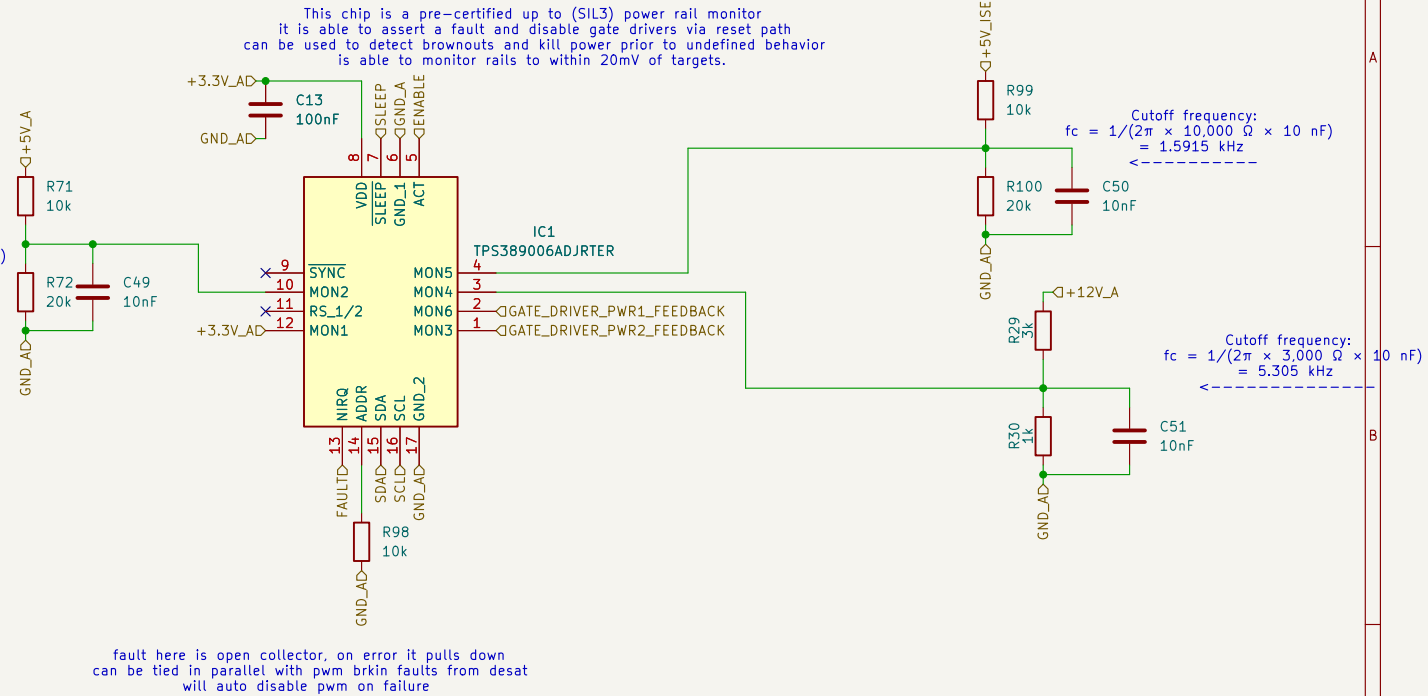
KiCad E.D.A. 10.0.3

Rev: A

Id: 10/21



Cutoff frequency:
 $f_c = 1/(2\pi \times 10,000 \Omega \times 10 \text{ nF})$
 $= 1.5915 \text{ kHz}$



HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/

File: PowerSupplyMonitor.kicad_sch

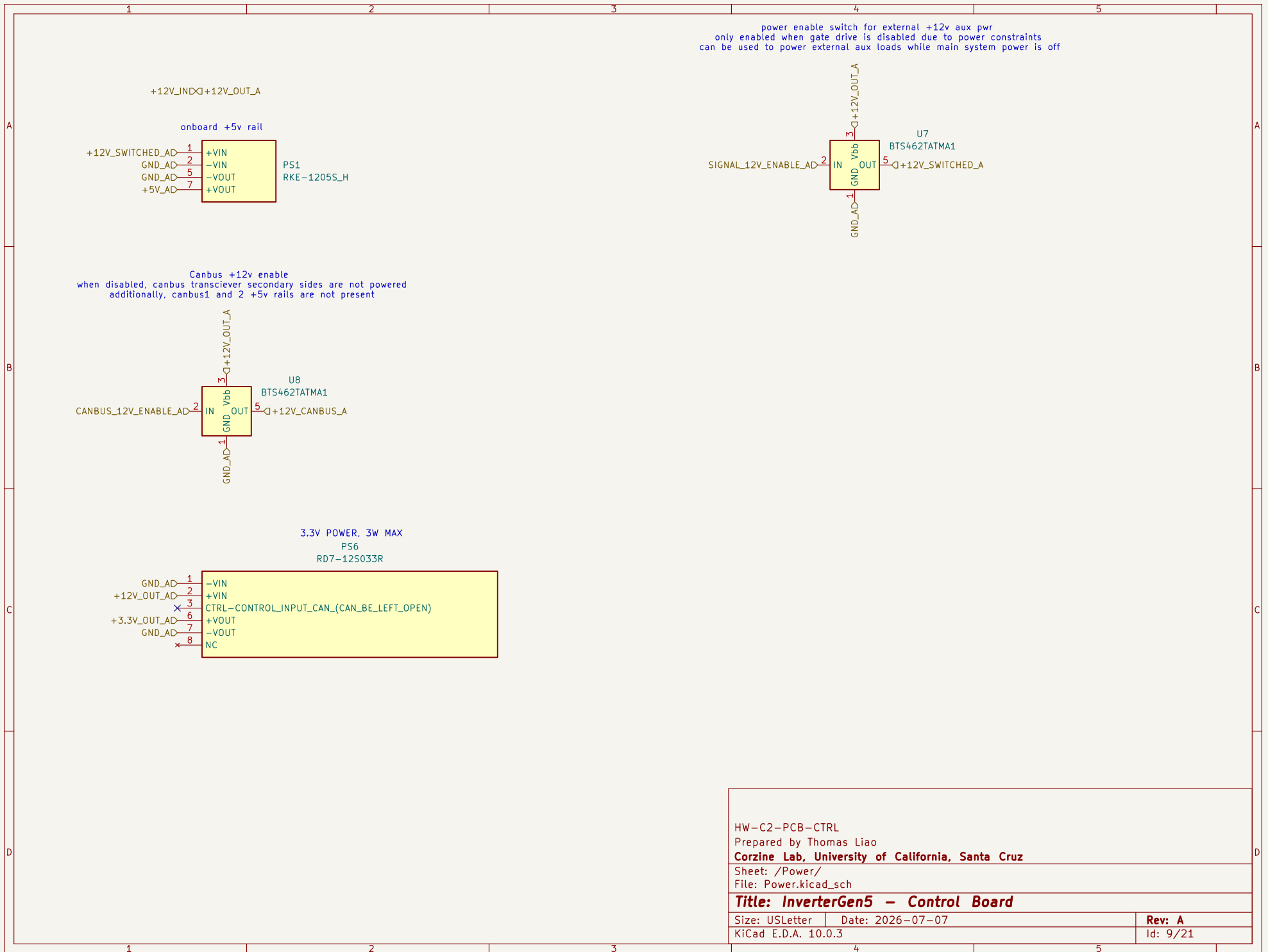
Title: InverterGen5 - Control Board

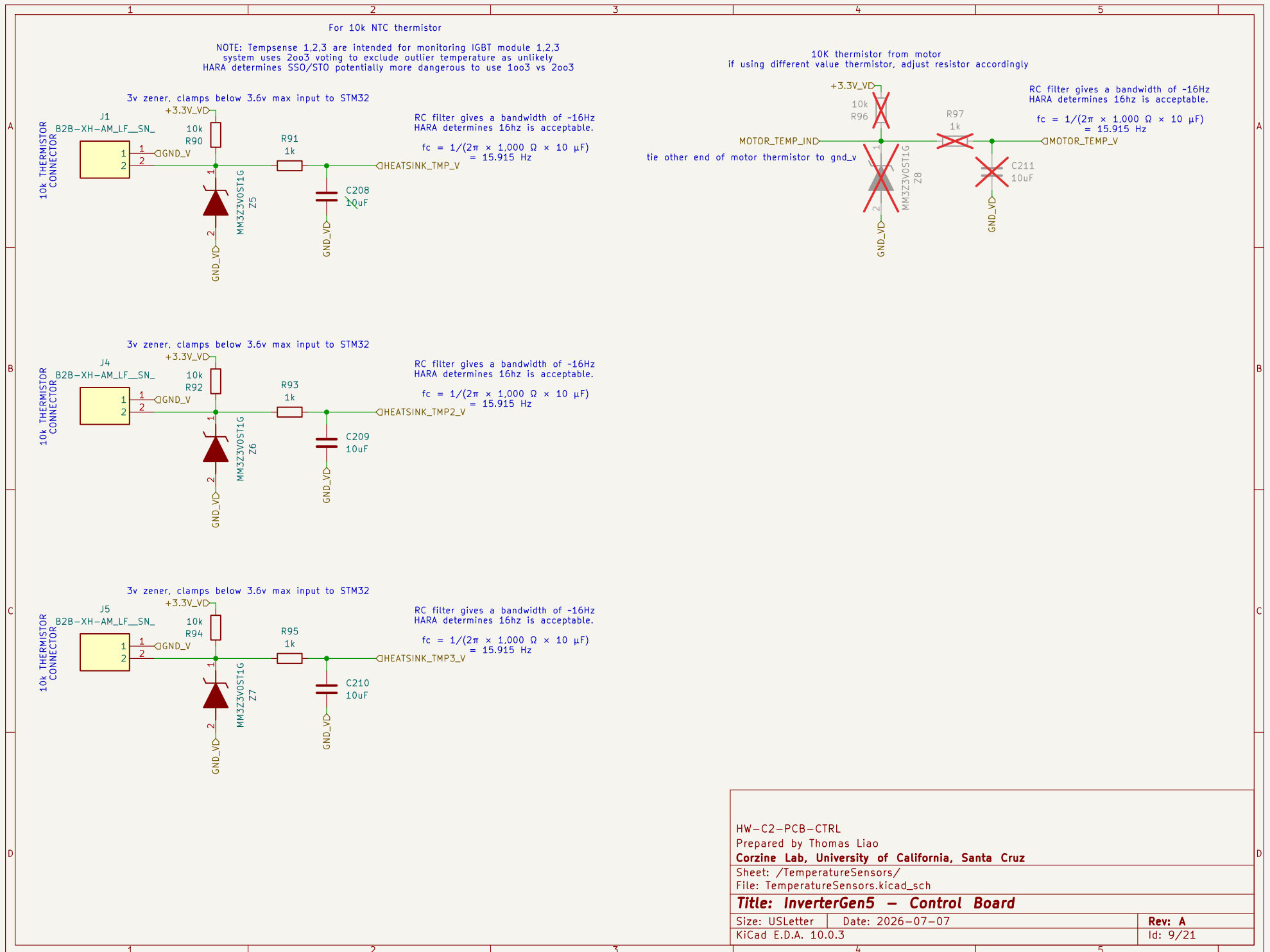
Size: USLetter Date: 2026-07-07

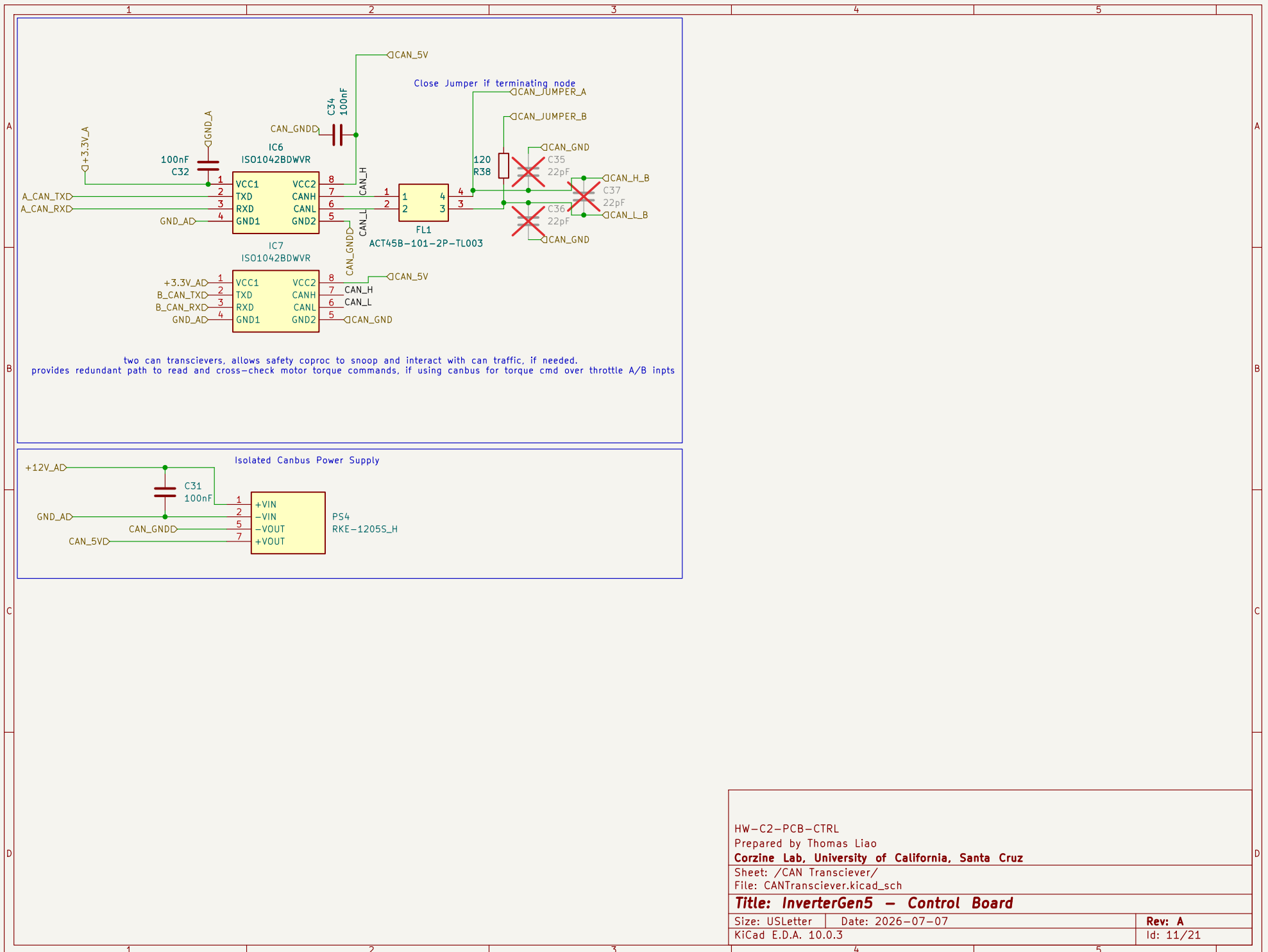
KiCad E.D.A. 10.0.3

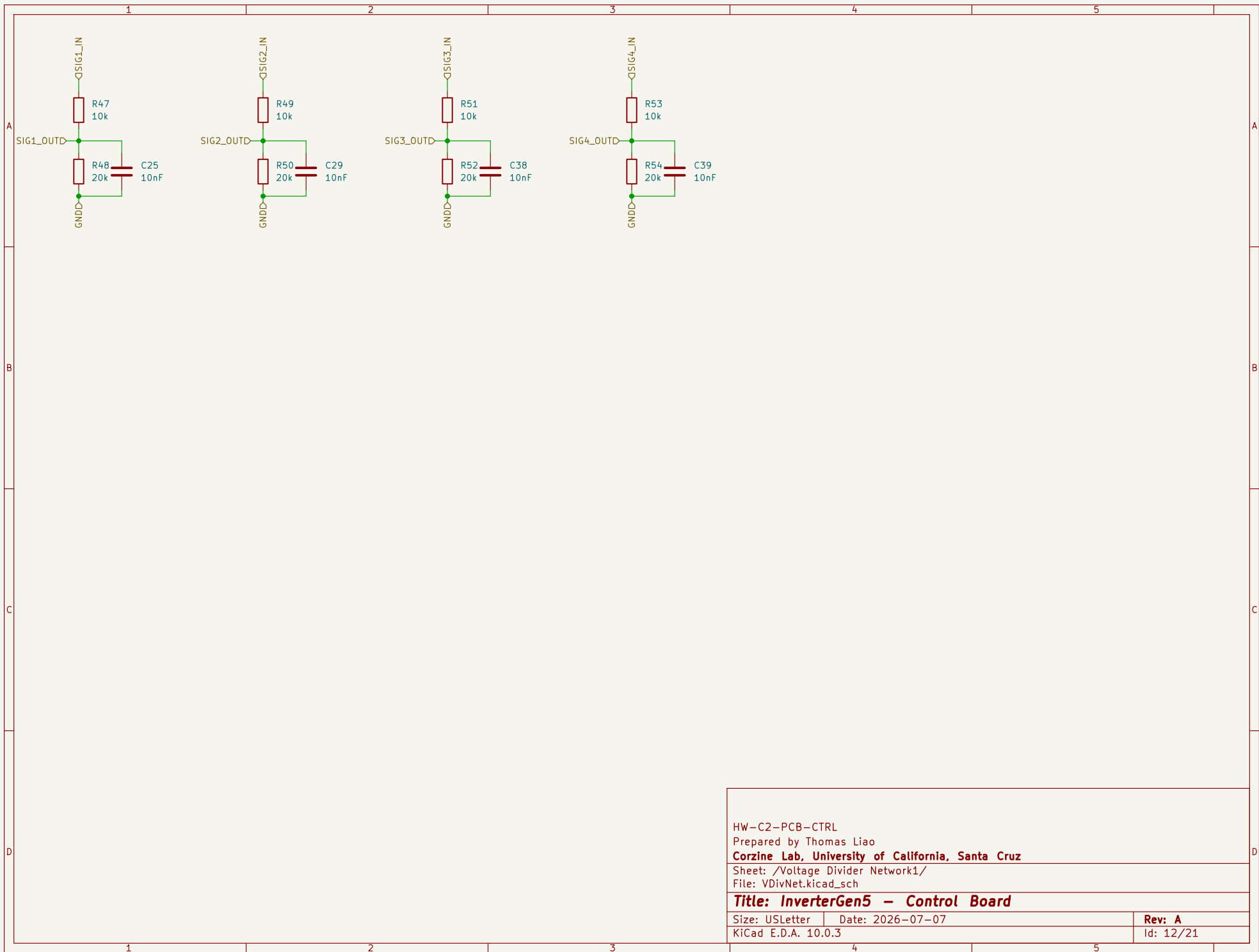
Rev: A

Id: 20/21









HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/

File: VDivNet.kicad_sch

Title: InverterGen5 - Control Board

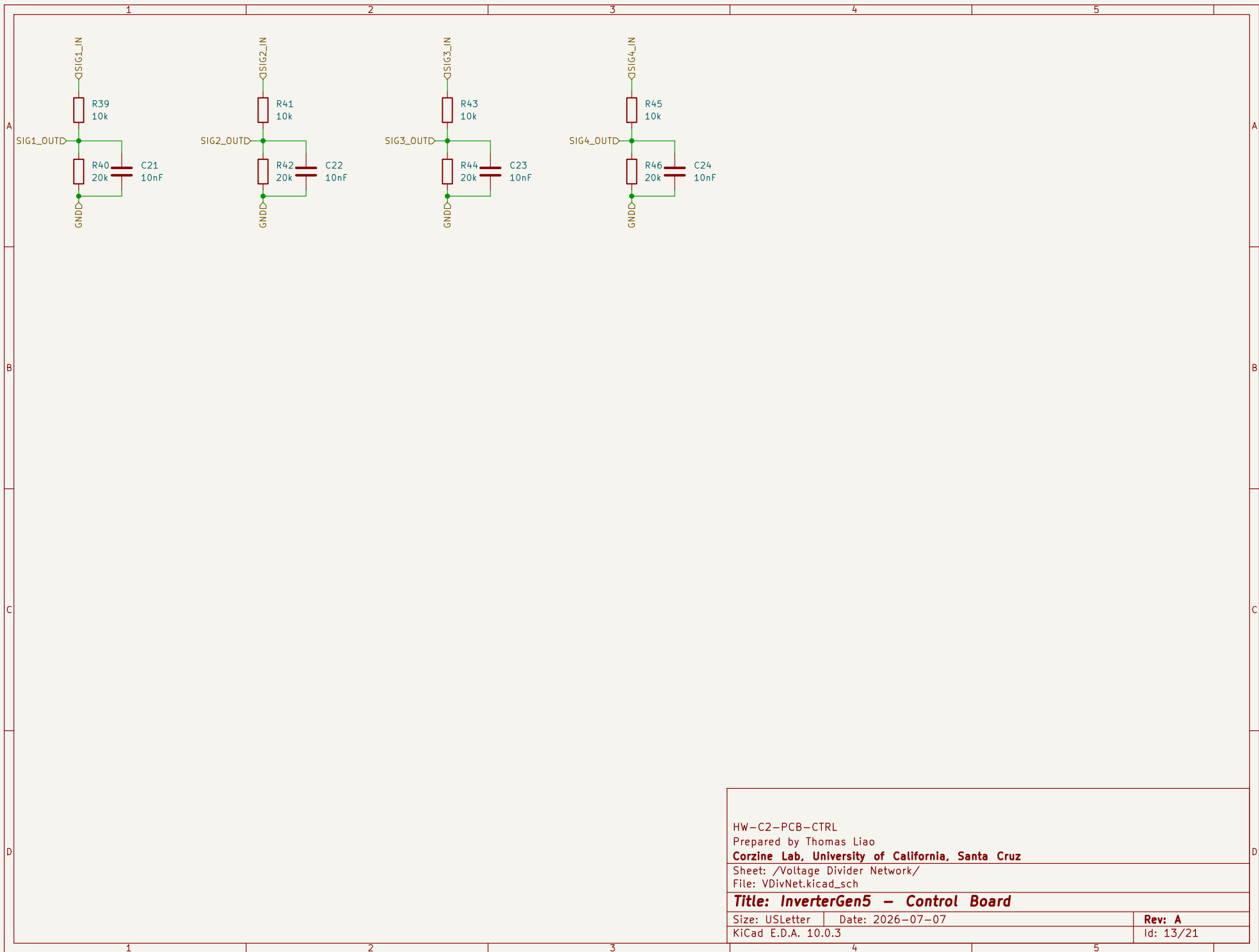
Size: USLetter

Date: 2026-07-07

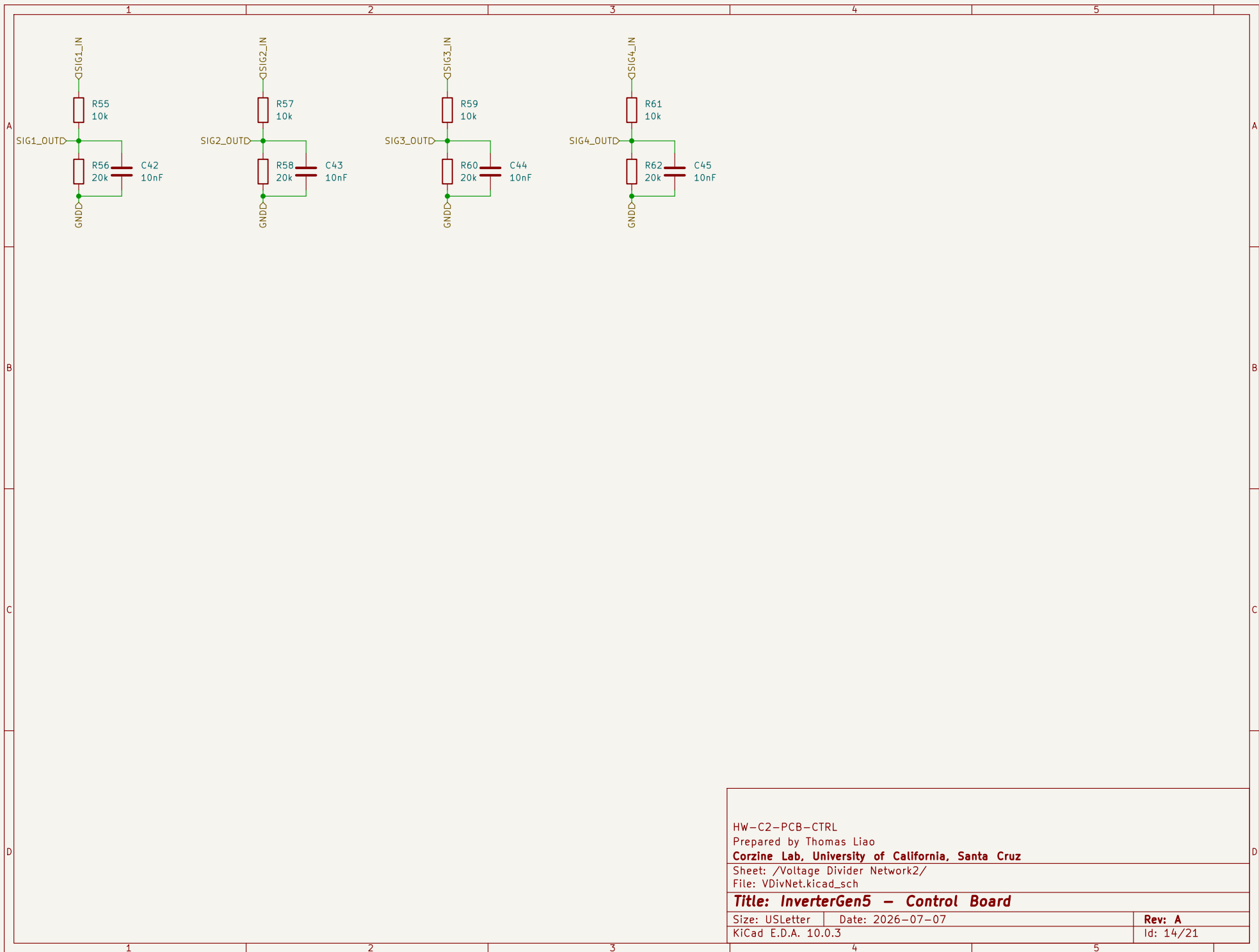
Rev: A

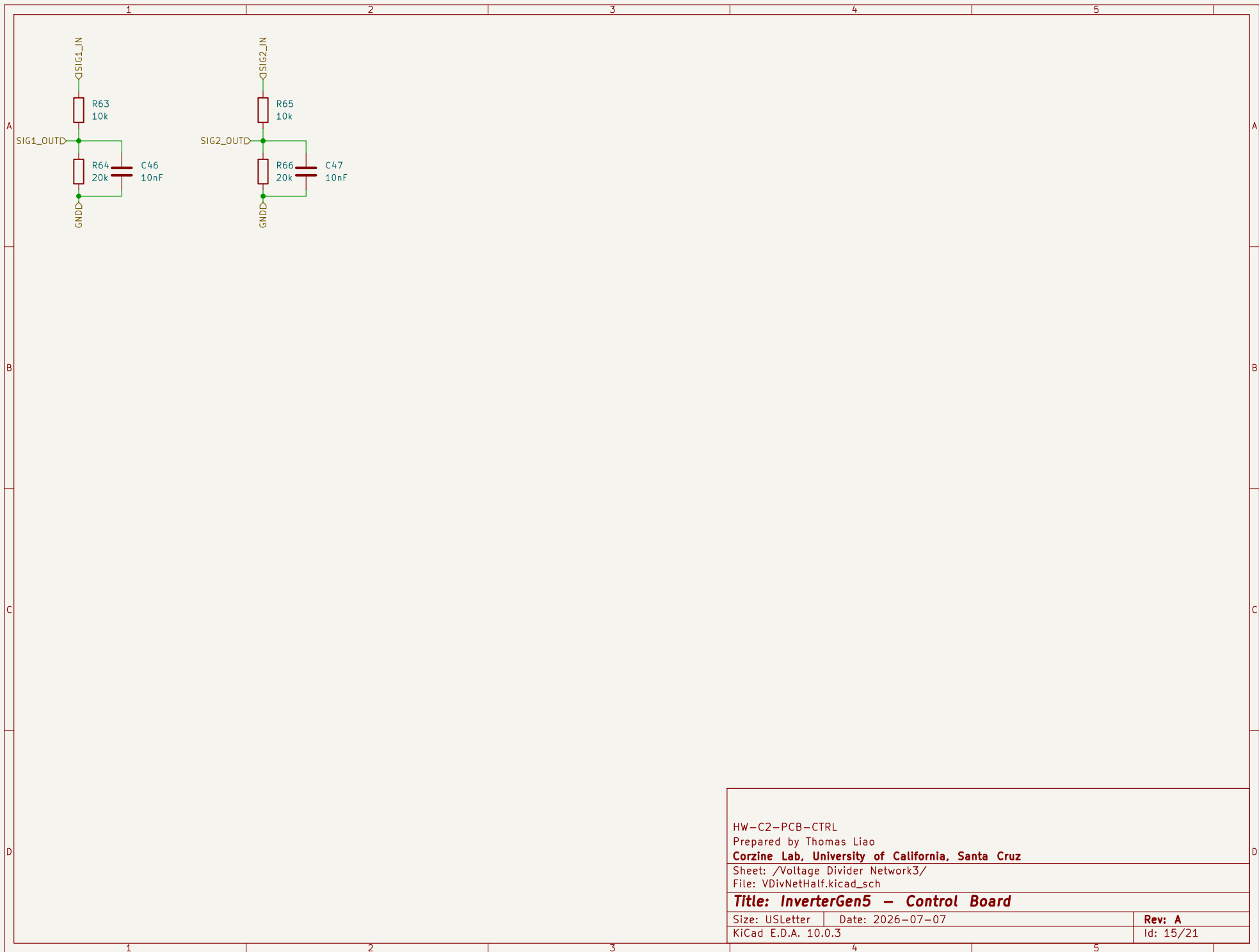
KiCad E.D.A. 10.0.3

Id: 12/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch
Title: InverterGen5 - Control Board
Size: USLetter | Date: 2026-07-07 | Rev: A
KiCad E.D.A. 10.0.3 | Id: 13/21





HW-C2-PCB-CTRL		
Prepared by Thomas Liao		
Corzine Lab, University of California, Santa Cruz		
Sheet: /Voltage Divider Network3/		
File: VDivNetHalf.kicad_sch		
Title: InverterGen5 - Control Board		
Size: USLetter	Date: 2026-07-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 15/21

